A Century Is Not Enough: Certification Bounds for Out-of-Sample Return Predictability

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Abstract

How precisely can a century-scale equity-premium dataset constrain economically meaningful out-of-sample predictability? We first state a minimum detectable effect under an explicit Gaussian constant-effect benchmark: for a one-sided test at size α and target power π, the smallest annualised information ratio detectable from T years of *evaluation* is $[z_{1-\alpha} + z_\pi]/\sqrt{T}$. This is a model-based power calculation, not a distribution-free impossibility bound. Because the design reserves twenty years for initialisation, the 1926–2025 data give at most eighty years of out-of-sample evaluation, so the detectable ratio is 0.28 for one prespecified predictor and 0.44 after Bonferroni correction over the forty-six candidates Goyal et al. [1] examine—against Campbell and Thompson's economic-significance threshold of 0.246. The design is short by twenty-three years for a single test and by one hundred and seventy-three once the search is priced. We then apply a drift-robust, anytime-valid certificate to 117 predictor–frequency pairs. Under the primary specification no pair reaches the individual 5% threshold and none survives e-BH; across nine design specifications applied to the fourteen classic monthly predictors, none does either. Among the 116 unflagged pairs the median of the *marginal* 95% upper endpoints is 0.60 and 112 exceed 0.25, so most intervals remain compatible both with no incremental predictability and with effects conventionally called economically meaningful. Sequential paths show the accumulation is highly uneven across predictors and across time: of six classic predictors, only inflation accumulates evidence in both halves of the window, the Treasury-bill rate only in the second, the term spread gains in the first and surrenders it in the second, and the three valuation ratios accumulate in neither. These describe when the procedure gathered evidence and are not estimates of structural-break dates.

1 Introduction

The out-of-sample predictability literature asks whether a variable forecasts the equity premium. After Welch and Goyal [2] the consensus answer is mostly no, and after Goyal et al. [1] it is mostly no again for a further twenty-nine candidates. This paper asks a prior question that determines what any such answer can mean: *given a sample of a certain length, what size of effect could it have detected at all?*

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Preprint. Code, pinned data, and one script per reported number: https://github.com/ITheClixs/crypto-return-predictability

Under an explicit Gaussian constant-effect benchmark the question has a closed-form answer, and it is uncomfortable. Section 3 derives a minimum detectable effect: the smallest annualised information ratio a design evaluating T years can detect at level α with power π. It is a power calculation under a stated model, not a distribution-free impossibility theorem, and Remark 2 says so precisely. Two features of it matter. The floor scales with $1/\sqrt{T}$ in *calendar span* and is almost unaffected by sampling frequency, so a century of monthly data and a century of annual data face nearly the same bound despite a twelvefold difference in observations. And it degrades with the size of the search, which in this literature is large and known.

Put those together against the effect the literature itself treats as meaningful—Campbell and Thompson [3] take a monthly out-of-sample R^2_{OS} of 0.5%, an annualised information ratio of 0.246, as economically significant—and the arithmetic is stark. Under the Gaussian benchmark of Section 3, a single predictor needs 103 years of *evaluation* to reach 80% power against it, and 253 once the 46-predictor search Goyal et al. [1] ran is priced. The canonical dataset begins in 1926 and the design reserves its first twenty years, leaving eighty.

What we do with that. A bound on what a study could have found is only useful if paired with a bound on what it did find. Section 4 uses an anytime-valid *certificate* [4], a non-negative wealth process whose validity is a finite-sample theorem and whose logarithm is the profit and loss of an explicit drift-hedged strategy. Two properties earn its place here. It supplies a time-uniform *ceiling* on a predictor's incremental value, which is the number this literature has never reported and the one a reader can act on. And because evidence compounds rather than averages, its path records *when* evidence arrived: a covariance averaged over a century weights an early contribution and a late one identically, whereas a compounding process does not. We read the path as a description of the procedure's evidence accumulation, not as an estimate of a break date.

What we find. Across 117 predictor–frequency pairs from 1926 to 2025, nothing is certified, under any of nine specifications of burn-in, window and payoff. That is unsurprising and is not the contribution. The contribution is the interval: across the 116 unflagged pairs the median of the *marginal* 95% upper endpoints on the incremental annualised information ratio is 0.60, and 112 of them sit above Campbell and Thompson's 0.25. These are pairwise bounds, not a simultaneous statement about a median parameter. Read as such they say that for almost every predictor in the set, the data rule out neither zero nor an economically meaningful edge.

The timing result is secondary and needs stating carefully. Across all 117 pairs, Clark–West statistics and certificate e-values are *positively* rank correlated, at a Spearman coefficient of $+0.54$ ($+0.55$ over the 116 unflagged). The two are measuring overlapping things, as they should. Among six classic predictors we display for illustration they disagree sharply, at -0.83; that subset shows the two *can* diverge and why, not that they generally invert. Read on the certificate's underlying process, the three valuation ratios—the predictors Clark–West scores highest, at 1.34 to 1.64—accumulate no evidence in either half of the window, while inflation, the Treasury-bill rate and the term spread do, in different halves. A statistic that averages a covariance over a century weights the two halves equally; a process that compounds does not, and the difference is visible.

What this is not. We do not claim the certificate is more powerful than Clark–West. It is not: Guven [4] measures the exchange rate at roughly a factor of two in sample size, the price of validity at every stopping time and under weaker assumptions. Nor do we claim to overturn Welch and Goyal [2]; we agree with it and reproduce it. The claim is narrower and, we think, more useful: this literature has been operating at or below the detection threshold of its own data for its entire history, the threshold is computable in advance, and the quantity worth reporting is a ceiling rather than a p-value.

2 Related work

The predictability literature and its instability. Welch and Goyal [2] evaluate seventeen predictors of the equity premium and find that essentially none beats the historical mean out of sample; Goyal et al. [1] repeat the exercise for twenty-nine further candidates published since, and reach the same verdict for most of them. Campbell and Thompson [3] argue that weak predictability can still be economically meaningful once forecasts are constrained by economic priors, and set the benchmark effect size we use throughout. Rapach et al. [5] recover out-of-sample gains from forecast combination. Running underneath all of it is instability: the same predictor works in one era and not the next, which is the observation the sequential paths of Section 6 illustrate and Proposition 4 prices.

Power, and what a test can be expected to find. Inoue and Kilian [6] show that out-of-sample tests are typically *less* powerful than in-sample ones, because splitting a finite sample costs information, and warn against reading out-of-sample failure as evidence that in-sample evidence was spurious. Our contribution is complementary and more absolute: their comparison is between two procedures on the same data, whereas Proposition 1 asks what a span of a given length can support at a stated size and power under a stated model—a design question rather than a procedure comparison. It is not a statement about all procedures, and Remark 2 draws that line. Hansen et al. [7] address a related selection problem by constructing a set of models that cannot be distinguished from the best. Where a model confidence set answers "which of these can be told apart", we ask "could any of them have been told apart from nothing, given this much data". We also run their procedure on our own forecasts in Section 6.1, where it retains every model including the benchmark.

Nested forecast comparison. Diebold and Mariano [8] is not valid for nested models; Clark and McCracken [9] and Clark and West [10, 11] supply the corrections this literature uses, and Guven [4] shows that the benchmark against which the correction is applied determines what null is being tested. We use Clark–West throughout as the incumbent and do not propose to replace it.

Game-theoretic statistics. The certificate is built from test martingales and e-values [12–15], with merging rules from Vovk and Wang [16] and false-discovery control from Wang and Ramdas [17]. Applications of betting arguments to finance exist—Kumon et al. [18] test whether observed *price processes* are martingales using limit-order betting strategies—but that is a specification test of the price process rather than a comparison of a forecasting model against an estimated benchmark, and it does not treat the drift as a nuisance. We are not aware of prior anytime-valid analysis of equity-premium predictability.

3 The certification bound

Let a signal-weighted strategy have annualised information ratio IR, and consider testing whether $\mathrm{IR} > 0$ from T years of data at level α.

Proposition 1 (Gaussian minimum detectable information ratio)**.** *Let $X_1, \ldots, X_n$ be independent Gaussian strategy returns with mean μ and known variance σ^2, let there be P observations per year, let $n = PT$, and define the annualised information ratio $\rho = \sqrt{P}\,\mu/\sigma$. For the one-sided test of $H_0 : \mu \leq 0$ against $H_1 : \mu > 0$ at size α, the uniformly most powerful test rejects when $\sqrt{n}\,\bar{X}/\sigma \geq z_{1-\alpha}$, and its power is*

$$\pi(\rho, T) \;=\; \Phi\big(\rho\sqrt{T} - z_{1-\alpha}\big).$$

The smallest annualised ratio attaining target power π from T years of evaluation is therefore

$$\rho_{\mathrm{MDE}}(T, \alpha, \pi) \;=\; \frac{z_{1-\alpha} + z_\pi}{\sqrt{T}},$$

and the span needed against a given ρ is $[z_{1-\alpha} + z_\pi]^2/\rho^2$. For m prespecified hypotheses under Bonferroni, replace α by α/m.

We write *floor* for ρ_{MDE} throughout, as shorthand for "the minimum detectable ratio of this design at this size and power". It is a property of a design, not of the class of all procedures; Remark 2 states the difference and we do not use the word in the stronger sense anywhere.

Remark 2 (What this is and is not)**.** Proposition 1 is a design-stage power calculation under a stated model, not a distribution-free lower bound over all procedures. Effects below ρ_{MDE} can still produce rejection; they do so with probability below π. The calculation assumes independent Gaussian returns, a constant effect, known variance, and a strategy fixed in advance; with serial dependence σ^2 must be replaced by a long-run variance, and with an estimated variance the finite-sample distribution is noncentral t rather than normal. A genuinely procedure-independent statement

Table 1: Minimum detectable annualised information ratio at 80% power, and the evaluation span needed against Campbell and Thompson's economic-significance threshold. Every entry comes from one script (`goyal_welch_mde.py`) and the same expressions. The threshold is reported unrounded at 0.2456: rounding it to 0.25 shortens the required span by about four years. The data span 100 calendar years, but the twenty-year initialisation leaves at most **80** years of out-of-sample evaluation, which is the span the test actually has.

Correction	α	MDE at 80 y	MDE at 100 y	Years needed for $\rho = 0.2456$
One predictor	0.0500	0.278	0.249	103
Bonferroni over 17 [2]	0.0029	0.402	0.360	214
Bonferroni over 46 [1]	0.0011	0.437	0.391	253
Bonferroni over 100	0.0005	0.462	0.413	283

Available evaluation span: 80 years. Short by 23 years for one test, 173 corrected.

would require a minimax or information-theoretic argument under a specified experiment, which we do not attempt. We use the benchmark because it is the calculation the applied literature would itself perform, not because it is the sharpest available.

Corollary 3 (Frequency invariance under square-root-of-time scaling)**.** *Under the assumptions of Proposition 1—independent increments, a constant effect, non-overlapping observations, and mean and variance scaling as P and $\sqrt{P}$ so that ρ is preserved—ρ_{MDE} depends on the calendar span T and not on the number P of observations per year.*

The cancellation is exact only under those assumptions. It need not hold under serial correlation, overlapping observations, conditional heteroskedasticity, recursive estimation, predictor persistence, or when lower-frequency series carry information that higher-frequency ones do not. Section 6 reports that the measured thresholds are numerically close across frequencies in this dataset, which is evidence that the approximation is adequate here and not a proof that it holds generally.

Proposition 1 assumes the effect is present throughout. It is not, in this literature, and the correction is unfavourable.

Proposition 4 (Intermittent effects raise the detectable threshold)**.** *Let $I_t \in \{0,1\}$ be an indicator, independent of the innovation, with $P(I_t = 1) = \phi$, and let $y_t = mI_t + \varepsilon_t$ with $\mathbb{E}[\varepsilon_t] = 0$ and $\mathrm{Var}(\varepsilon_t) = \sigma^2$. For a strategy sized on $g_t = mI_t$, the per-period payoff $X_t = g_t y_t$ has $\mathbb{E}[X_t] = \phi m^2$ and $\mathrm{Var}(X_t) = \phi m^2\sigma^2 + \phi(1-\phi)m^4$, so with $\rho_{\mathrm{on}} = m/\sigma$ the per-period ratio is*

$$\rho_{\text{measured}} = \frac{\sqrt{\phi}\,\rho_{\text{on}}}{\sqrt{1 + (1-\phi)\rho_{\text{on}}^2}} \approx \sqrt{\phi}\,\rho_{\text{on}} \qquad \textit{for small } \rho_{\text{on}},$$

and under that approximation the span required scales as $1/(\phi\,\rho_{\text{on}}^2)$: a factor $1/\phi$ longer than for an always-present effect of the same on-period strength.

The second denominator term is the variance contributed by switching between regimes, and it is not negligible in general: at a per-period ρ_{on} of 0.58 the approximation overstates the ratio by up to 14% across duty cycles, and at 1.7 by up to 90%, in both cases worst at the smallest ϕ. At the magnitudes this paper uses it is negligible—an annualised on-period ratio of 0.5 is a per-period 0.14 at monthly frequency, where the error stays under 1% for every ϕ. We confirm the exact expression against four million simulated periods at each of seven duty cycles and three ratios; the largest discrepancy is 0.99% and occurs where Monte Carlo noise dominates. The strategy also has to be able to size on I_t, which requires the regime to be known in advance; where it is not, the achievable ratio is lower still, so the approximation errs in the direction that understates the problem.

Remark 5 (We do not estimate ϕ from the wealth path)**.** It is tempting to read a duty cycle off the interval between the start of evaluation and the year the certificate's wealth peaks, and an earlier draft did. We do not. The peak is a random, post-hoc quantity that depends on noise, stake calibration and the payoff; it is not a changepoint estimator and carries no interval. Where a duty cycle appears below it is measured as the fraction of equal sub-periods with positive accumulation—a descriptive statistic—or supplied externally as a hypothetical.

Table 1 states the position, and the distinction between the raw and the evaluation span matters. Against the raw hundred years a single test would sit within three years of the boundary; against the

Table 2: The same effect in four currencies. Monthly out-of-sample R^2_{OS} as reported by the literature; the implied per-period and annualised information ratios; and the calendar span a one-sided fixed-sample test needs to reach 80% power against it, uncorrected and Bonferroni-corrected for the 46-predictor search of Goyal et al. [1].

Monthly R^2_{OS}	Per-period IR	Annualised IR	Years, one test	Years, 46 tests
0.1%	0.032	0.11	515	1271
0.2%	0.045	0.16	257	635
0.5%	0.071	0.25	103	253
1.0%	0.101	0.35	51	126
2.0%	0.143	0.49	25	62

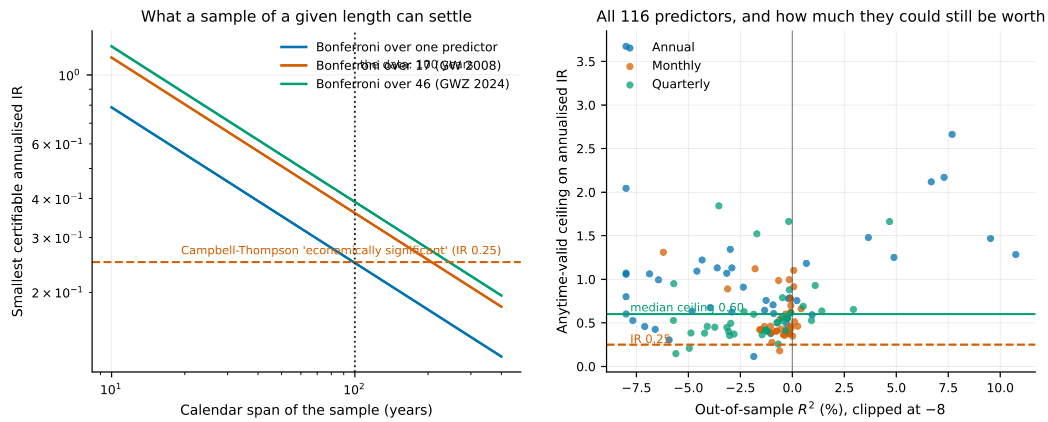

Figure 1: **Left:** the minimum detectable ratio against evaluation span, for searches of one, seventeen and forty-six predictors. The horizontal line is Campbell and Thompson's economic-significance threshold; the vertical line is the data. **Right:** every predictor–frequency pair, plotted as its anytime-valid ceiling against its out-of-sample R^2_{OS}. The median of the marginal upper endpoints is 0.60 and 112 of 116 lie above 0.25: for almost every predictor the data rule out neither zero nor an economically meaningful edge.

eighty years the design actually evaluates it is short by twenty-three, with no margin for parameter instability, transaction costs, or a slightly smaller true effect. Corrected for the search that was performed, the shortfall is one hundred and seventy-three years.

Reading the floor in the units the literature uses. Table 2 converts a monthly out-of-sample R^2_{OS}—the quantity this literature reports—into the annualised information ratio the bound is stated in, and then into required span. The row that matters is the third: Campbell and Thompson's threshold for economic significance corresponds to an annualised ratio of 0.25 and needs 103 years for a single predictor, 253 once the search is corrected for. Even a monthly R^2_{OS} of 1%, twice what that literature treats as meaningful and larger than anything in our results, needs 126 years under the correction.

4 The instrument

We use the drift-robust certificate of Guven [4]. Because every number in Section 6 depends on it, we state the construction in full rather than by reference.

Setup. Let $\mathcal{F}_t$ be the filtration generated by everything observable through the close of period t. The outcome y_t is the period-t log equity premium; the signal g_t is the period-t forecast from the predictive regression minus the intercept-only forecast from the *same* estimation window, so g_t is $\mathcal{F}_{t-1}$-measurable and equals zero exactly when the two forecasts agree. Write $\hat{\sigma}_t$ for the running standard deviation of the outcome through $t-1$ and $z_t = g_t/\hat{\sigma}^g_t$ for the signal standardised by its own running scale, again using only $\mathcal{F}_{t-1}$. The null is

$$H_0: \quad \mathbb{E}[\, y_t - \mu \,|\, \mathcal{F}_{t-1}] = 0 \quad \text{for some fixed but unknown } \mu \in \mathbb{R},$$

that is, the outcome is a martingale difference around *some* drift and the signal carries no information about it. The drift μ is a nuisance and is not assumed known.

Two martingales. Fix a candidate drift μ and a payoff φ; throughout we take $\varphi(u) = u$, the identity, which requires an a-priori envelope B with $|y_t| \le B$ and nothing else. Put $\psi_t(\mu) = \varphi((y_t - \mu)/\hat{\sigma}_t)$ and let $E_t = \sup_u |\varphi(u)|$ over the reachable range be the corresponding payoff envelope. Define

$$W_t^{\mathrm{sig}}(\mu) = \prod_{s \le t} \big(1 + \lambda_s \, g_s \, \psi_s(\mu)\big), \qquad \lambda_s \in \Big[0, \ \tfrac{c}{|g_s| E}\Big] \text{ predictable,}$$

$$W_t^{\mathrm{ctr}}(\mu) = \tfrac{1}{2} \prod_{s \le t} \big(1 + \kappa \psi_s(\mu)\big) + \tfrac{1}{2} \prod_{s \le t} \big(1 - \kappa \psi_s(\mu)\big), \qquad \kappa = \tfrac{c'}{E},$$

with $c = 0.9$ and $c' = 0.9$ so that every factor stays strictly positive. The first is the wealth of a bettor who stakes a predictable fraction on the signal leading the outcome; the second is the average of two bettors staking in opposite directions on μ being the wrong centre. The stakes λ_s are chosen by the usual predictable-mixture rule from the realised payoffs through $s - 1$, capped at $c/(|g_s| E)$; because they are $\mathcal{F}_{s-1}$-measurable, no choice of λ can affect validity, only power.

The certificate. With $w = 0.9$,

$$\mathcal{E}_t^{\mathrm{raw}} \; = \; \inf_{\mu \in \mathbb{R}} \Big[\, w \, W_t^{\mathrm{sig}}(\mu) + (1 - w) \, W_t^{\mathrm{ctr}}(\mu) \, \Big], \qquad \mathcal{E}_t \; = \; \max_{s \le t} \, \mathcal{E}_s^{\mathrm{raw}}.$$

Under H_0 each of $W_t^{\mathrm{sig}}(\mu^\star)$ and $W_t^{\mathrm{ctr}}(\mu^\star)$ is a non-negative martingale at the true drift $\mu^\star$, hence so is their convex combination, and an infimum over μ is bounded above by the value at $\mu^\star$. Ville's inequality gives $P\big(\exists t : \mathcal{E}_t \ge 1/\alpha\big) \le \alpha$ in finite samples, uniformly in time and uniformly in the unknown drift. At a wrong μ the centring bettor gets rich, so the infimum is not dragged to zero by candidates the data have already ruled out; the nuisance is *eliminated* rather than estimated, which is what removes the bootstrap.

Two numerical points matter and we state them because getting either wrong is silent. The infimum is computed on a grid, and the grid spacing must resolve the drift to better than its sampling error $\hat{\sigma}/\sqrt{n}$—a grid spaced at $\hat{\sigma}$ produces an infimum above the true one and leaks type-I error. And $\mathcal{E}_t$ is the running maximum, which is the object Ville's inequality bounds and the only one a test may use; $\mathcal{E}^{\,\mathrm{raw}}$ is the process itself and is the object any *diagnostic* must use, since a running maximum rises only when a new high is set. Table 4, Figure 2 and Section 6.3 all use $\mathcal{E}^{\,\mathrm{raw}}$.

From a certificate to a ceiling. Inverting the same construction over a value parameter, rather than testing it against zero, gives a time-uniform confidence interval for

$$\theta \; = \; \mathbb{E}\big[z_t \, (y_t - \mu)\big],$$

the expected per-period return of a position sized by the standardised signal. For each candidate (θ, μ) a two-sided betting martingale tests $\mathbb{E}[z_t(y_t - \mu)] = \theta$; it is averaged with the centring martingale as above, and θ is retained when the infimum over μ never reaches $1/\alpha$. The retained set is an interval and its endpoints are found by bisection. The following converts its upper endpoint into the units this literature uses.

Lemma 6 (Value to annualised information ratio)**.** *Let σ_y be the outcome's standard deviation and P the number of periods in a year. A position taking z_t units of the outcome earns θ per period in expectation with per-period standard deviation $\sigma_y \sqrt{\mathbb{E}[z_t^2]}$, so its per-period information ratio is $\theta/(\sigma_y \sqrt{\mathbb{E}[z_t^2]})$. The standardisation makes $\mathbb{E}[z_t^2] \to 1$, and under serially uncorrelated payoffs the ratio scales with the square root of the aggregation, giving*

$$\rho \; = \; \frac{\sqrt{P}\, \theta}{\sigma_y}.$$

Applying this to the interval's upper endpoint gives the reported ceiling.

Lemma 7 (Out-of-sample R^2_{OS} to information ratio)**.** *Let the benchmark forecast have mean-squared error σ_y^2 and let the model's incremental forecast be g_t with $\mathbb{E}[g_t^2] = \sigma_g^2$. Then $R^2_{OS} = 1 - \mathbb{E}[(y_t - \hat{y}_t)^2]/\mathbb{E}[(y_t - \bar{y})^2] = \sigma_g^2/\sigma_y^2$ when the incremental forecast is orthogonal to the benchmark error, and a position sized proportionally to g_t has per-period ratio $\sigma_g/\sigma_y = \sqrt{R^2_{OS}}$. Annualising,*

$$\rho \;=\; \sqrt{P\, R^2_{OS}}.$$

At $P = 12$ and $R^2_{OS} = 0.5\%$ this is $\rho = 0.2450$; the 0.2456 used in Table 1 comes from the same calculation applied to the log-premium series' own moments.

Both conversions assume serially uncorrelated payoffs and unit-scale standardisation; neither is exact and both are stated so a reader can check them against their own convention. They are the only place a squared-error number and a betting number are made comparable, and every such comparison in this paper routes through them.

Multiplicity without a bootstrap. E-values average to an e-value under arbitrary dependence, so the grid-level statistic over the 117 dependent tests is their mean, and false-discovery control follows from e-BH [17] with no assumption on the dependence structure and no resampling.

5 Data and design

We use the Goyal et al. [1] dataset through December 2025: monthly, quarterly and annual equity-premium series from 1926 together with the seventeen classic Welch and Goyal [2] predictors and the twenty-nine added by Goyal et al.. The outcome is the log equity premium $\log(1 + r_t) - \log(1 + r_t^f)$; predictors are lagged one period; forecasts come from an expanding-window OLS regression with an intercept, re-estimated every period, and the benchmark is the same window's historical mean, which is that regression's intercept-only restriction. Following Goyal et al., the out-of-sample period begins twenty years after the in-sample period starts.

Real-time versus revised series. Many of the new predictors require re-estimation every period, and Goyal et al. ship them as vintage matrices whose rows index the period and whose columns index the vintage. The admissible series is the diagonal—what a forecaster actually had—and the revised series is the last column. We use the diagonal throughout and record the source per predictor. On the twenty-five predictors where both exist, the revised series looks *better* for only four, so this choice does not drive the results; it is nonetheless the difference between an out-of-sample claim and a claim that quietly uses full-sample parameter estimates.

Validation. Because everything here depends on the pipeline being right, we validate it against Goyal et al.'s published Table 3 Panel A, predictor by predictor, matching their evaluation window and their Campbell–Thompson restriction. Across the fourteen classic monthly predictors the correlation is 0.968 (Spearman 0.959), signs agree in 13 of 14, and the median absolute gap is 0.075 percentage points. The single disagreement is tms, where both values round to zero.

Remark 8 (The Campbell–Thompson restriction is not free)**.** Truncating the forecast at a non-negative equity premium helps 13 of the 14 predictors and moves the mean out-of-sample R^2_{OS} from -0.511% to -0.281%. Since the effects under discussion are themselves a few tenths of a percentage point, an economically motivated truncation is doing about as much work as any predictor in the set. It helps most exactly where the unrestricted regression produces the wildest forecast. We report unrestricted numbers by default.

6 Results

Two grids, kept separate. The *primary* grid is the 117 predictor–frequency pairs of Table 8, run once under the pre-specified design of Section 5. The *sensitivity* grid is a smaller object: the fourteen classic monthly predictors, re-run under nine designs. The two are reported separately throughout, because a maximum over 117 pairs and a maximum over 14 are not comparable quantities and an earlier draft quoted one for the other.

Table 3: Specification sensitivity. Scope is the *fourteen classic monthly predictors*, not the 117-pair primary grid; "Max e-value" is therefore a maximum over fourteen. Row two is the primary design restricted to those fourteen. Rejecting needs an e-value of 20.

Burn-in	Window	Payoff	Median R^2_{OS} (%)	Max e-value	Grid e-value	Certified
10 y	expanding	identity	-0.329	3.05	1.48	0
20 y	expanding	identity	-0.378	5.71	2.04	0
30 y	expanding	identity	-0.342	8.02	2.55	0
40 y	expanding	identity	-0.308	6.27	1.94	0
20 y	rolling 20 y	identity	-0.989	5.50	1.81	0
20 y	rolling 30 y	identity	-0.579	8.04	2.73	0
20 y	rolling 50 y	identity	-0.515	7.58	2.17	0
20 y	expanding	tanh	-0.378	1.74	1.24	0
20 y	expanding	sign	-0.378	1.38	1.08	0

Nothing is certified, and not for want of trying. On the primary grid, out-of-sample R^2_{OS} is negative for 95 of 117 (81%), no pair reaches the $\alpha = 0.05$ threshold, the largest e-value observed is 7.47, the grid-level merged e-value is 1.79 against the 20 required, and nothing survives e-BH. On the sensitivity grid, Table 3 sweeps burn-ins of ten to forty years, rolling windows of twenty to fifty years against the expanding one, and two alternative payoffs; nothing certifies under any of the nine, and the largest e-value seen anywhere in that sweep is 8.04 (fourteen predictors, rolling thirty-year window). The rolling windows are the informative negative—Welch and Goyal's complaint is instability, so a window that forgets is the specification most likely to find something, and it does not.

The ceilings are the result. The right panel of Figure 1 plots every pair. The median of the marginal 95% upper endpoints on the incremental annualised information ratio is 0.60; 112 of 116 exceed 0.25. By frequency the medians are 0.46 monthly, 0.53 quarterly and 0.86 annual. Each endpoint has time-uniform coverage for its own predictor–frequency pair; their median summarises the collection and is *not* itself a 95% bound on a median parameter, which would require a simultaneous or order-statistic construction we do not attempt. Read correctly, the summary of this literature is not "the predictors do not work" but a statement with a number in it: *the pairwise intervals are wide, with a median upper endpoint near* 0.6*, and eighty years of evaluation does not narrow them below the threshold this literature calls economically meaningful.*

Frequency does not buy power. The certifiable floor is 0.38 monthly, 0.37 quarterly and 0.35 annually over the same span, across 1200, 400 and 100 observations respectively—Corollary 3, measured.

When the evidence arrived. Table 4 and Figure 2 report where in the evaluation window each certificate accumulated its log wealth, measured on the underlying process rather than on the running maximum. Clark–West averages a covariance over the century and so weights the two halves equally; the certificate compounds and does not. Over the full grid the two rank predictors similarly, at Spearman $+0.54$ across all 117 pairs and $+0.55$ across the 116 unflagged. The six shown here are chosen for familiarity and are the part of the grid where the two disagree most—their own rank correlation is -0.83—so they illustrate the mechanism and should not be read as the general pattern. The valuation ratios contribute negatively in both halves; inflation, the Treasury-bill rate and the term spread contribute in different halves: inflation in both, the Treasury-bill rate only in the second, and the term spread gaining 1.16 nats in the first and surrendering 0.95 of it in the second. The three valuation ratios are negative in both halves. On the underlying process d/p and d/y rise above one only briefly in the 1950s and e/p never does, so their reported "peak years" are high-water marks of paths that start at one and end below it rather than dates at which anything stopped—precisely the artifact that reading a running maximum produces. These are descriptions of an evidence path, not break-date estimates, and they carry no interval.

Sample accounting. Four counts appear in this paper and they are different objects. Table 5 states each once.

Table 4: Six illustrative predictors: Clark–West against the certificate, and where in the window the certificate accumulated. "Nats" are log-wealth contributed in each half of the evaluation period, measured on the underlying process, not on the running maximum $\mathcal{E}_t$. "Peak" is the year that process is highest; it is a descriptive feature of a noisy path, not a break-date estimate. These six are the part of the grid where the two procedures disagree most (Spearman -0.83 here against $+0.54$ over all 117 pairs). †: the process exceeds one only transiently, so the peak is a high-water mark of a declining path. "—": the process never exceeds one, and no peak is meaningful.

Predictor	Clark–West	e-value	Peak year	Nats, 1st half	Nats, 2nd half
d/y	1.64	1.14	1957†	−0.17	−0.15
e/p	1.55	1.00	—	−1.18	−0.18
tbl	1.44	3.48	2021	−0.11	+1.11
d/p	1.34	1.45	1957†	−0.01	−0.43
tms	0.78	3.92	1987	+1.34	+0.03
$infl$	0.83	5.71	1999	+0.59	+1.16

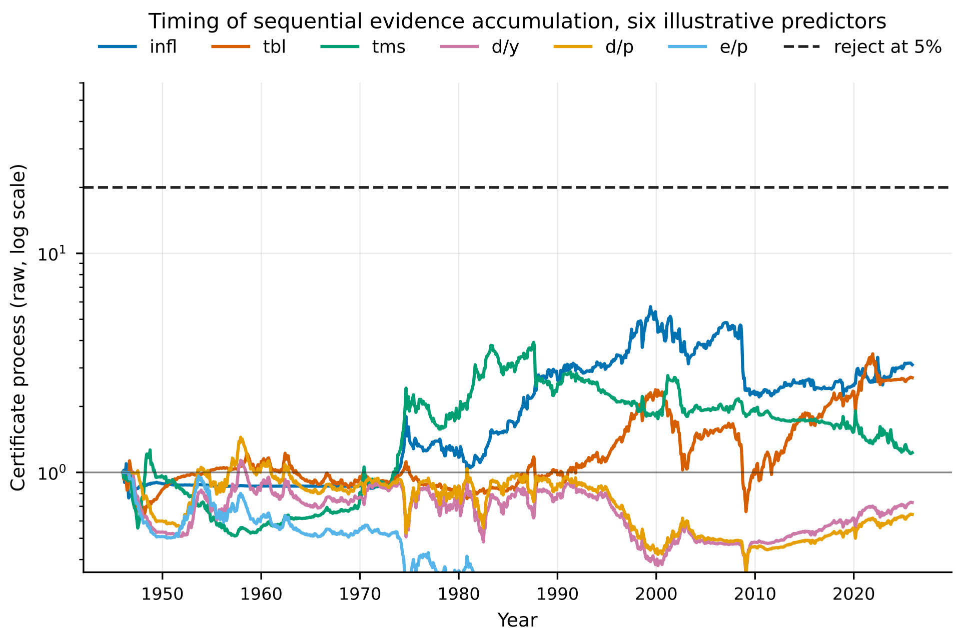

Figure 2: Certificate wealth for six illustrative predictors, plotted as the *underlying* process rather than the running maximum $\mathcal{E}_t$, so that periods of no accumulation appear flat or falling rather than merely level. Inflation, the Treasury-bill rate and the term spread accumulate, at different times and none decisively; the three valuation ratios do not. These are paths of a noisy stochastic process, not break-date estimates: they show when this procedure gathered evidence, and carry no interval on any change in the underlying process. Note that the three valuation ratios spend most of the window *below* one: d/p and d/y rise above it only transiently in the 1950s and e/p never does, so no durable evidence accumulates for any of them—a fact the running maximum conceals by construction. The six are the region of the grid where the certificate and Clark–West disagree most and are not representative of all 117 pairs.

6.1 Does the standard toolkit see anything we do not?

A non-result from a new instrument is worth little until the conventional instruments have been run on identical inputs. Table 6 does that on the fourteen classic monthly predictors, using the same walk-forward forecasts.

Diebold–Mariano on squared-error loss is reported for completeness and is *not* a valid test here: the comparison is nested, so under the null the loss differential is degenerate and the statistic is not asymptotically normal [9]. It rejects nothing, and it would be no evidence if it did. Clark–West, the

Table 5: What each count in this paper refers to.

Count	Object	Scope
117	Primary grid	Every predictor–frequency pair that clears the minimum-observation rule, run once under the pre-specified design. All rejection and merging statistics use this.
116	Unflagged pairs	The primary grid less the one degenerate fit (Section 6). Ceiling summaries use this, since a nine-sigma forecast makes a ceiling uninterpretable rather than wrong. Including it moves the median by less than 0.01.
56	Own-sample audit	Variables in Goyal et al.'s ReadMe with a recorded sample period from which a span can be computed; the audit is per *variable*, not per pair.
14	Sensitivity grid	The classic monthly predictors, re-run under nine designs.

Table 6: The conventional toolkit on the identical walk-forward forecasts, fourteen classic monthly predictors, 960 monthly observations each. DM is shown for completeness only and is invalid under nesting. ENC-t coincides with Clark–West by construction. No test rejects at 5% uncorrected, let alone corrected.

Predictor	DM (invalid)	p	Clark–West = ENC-t	p	In MCS
d/p	-0.16	0.56	1.34	0.090	yes
d/y	-0.73	0.77	1.64	0.051	yes
e/p	-0.73	0.77	1.55	0.060	yes
d/e	-1.33	0.91	-0.37	0.644	yes
b/m	-1.41	0.92	0.59	0.279	yes
$ntis$	-0.40	0.66	0.86	0.195	yes
tbl	0.09	0.47	1.44	0.075	yes
lty	-0.53	0.70	1.37	0.085	yes
ltr	-1.80	0.96	-0.56	0.713	yes
tms	-0.17	0.57	0.78	0.217	yes
dfy	-0.93	0.82	-0.75	0.774	yes
dfr	-0.87	0.81	-0.23	0.591	yes
$infl$	0.39	0.35	0.83	0.203	yes
$svar$	-0.63	0.74	-0.51	0.696	yes
Benchmark (historical mean) is also retained: 15 of 15 models in the 90% MCS.					

standard nested correction, rejects nothing at 5%; its largest statistic is 1.64 for d/y, a one-sided p of 0.051, which is not significant before any multiplicity correction and is far from it after. The Clark–McCracken ENC-t statistic is, for a one-step-ahead nested pair, *algebraically identical* to Clark–West: the adjusted loss differential is exactly twice the encompassing product and the factor cancels in the t-ratio. We verify this to machine precision in code and report it as one test rather than two, because presenting the same number under two names would overstate the corroboration.

The model confidence set of Hansen et al. [7], computed over the benchmark and all fourteen models jointly with a stationary bootstrap, retains *all fifteen* at the 90% level. That is the sharpest way to state the situation: after eighty years of monthly data, the historical mean cannot be distinguished from any predictor in the classic set, nor any predictor from any other. The MCS and the grid-level e-value answer the same question and give the same answer.

None of this makes the certificate right. It establishes the weaker thing the paper needs: the absence of a rejection is a property of the data at this sample size, not of our instrument, and the toolkit that produced the literature's positive results does not produce them here. What the toolkit does not supply, at any point, is a bound on how large the effect could still be. That is the gap the ceiling fills.

6.2 Could the samples ever have supported the claims?

The bound is a statement about a sample, so it can be applied retrospectively. Every variable in the set has a recorded data span, that span fixes the floor any test faces on it, and the comparison with a claimed effect is then arithmetic rather than opinion.

Table 7: The minimum detectable ratio implied by each variable's own sample span, at 80% power, for the twelve shortest-sampled. "Solo" is a single test at 5%; "corrected" applies Bonferroni over the 46-variable search. Both are compared against an annualised information ratio of 0.25.

Variable	Frequency	Span (years)	Floor, solo	Floor, corrected
rsvix	Monthly	29.6	0.46	0.72
impvar	Monthly	29.6	0.46	0.72
vp	Monthly	31.9	0.44	0.69
vrp	Monthly	34.9	0.42	0.66
crdstd	Quarterly	35.5	0.42	0.66
disag	Monthly	44.0	0.37	0.59
shtint	Monthly	52.9	0.34	0.54
sntm	Monthly	58.3	0.33	0.51
accrul	Annual	60.0	0.32	0.50
cfacc	Annual	60.0	0.32	0.50
csp	Monthly	65.6	0.31	0.48
ndrbl	Monthly	67.8	0.30	0.47
Median over all 56: span 99 y, solo 0.25, corrected 0.39				
Below 0.25: solo 33/56; corrected 0/56 under Bonferroni, Holm and Benjamini–Hochberg alike				

Table 7 does this for the 56 variables with a recorded sample. Of those, 23—41%—have a span too short to certify an annualised ratio of 0.25 even as a single test at the 5% level. Once the 46-variable search that generated the set is corrected for, the figure is 56 of 56. The median span is 99 years and the median corrected floor is 0.39, against a target of 0.25.

The result does not rest on a punitive correction. Bonferroni is the most conservative multiplicity adjustment in common use, so it is fair to ask whether the headline is an artifact of choosing it. It is not, for a structural reason: at the *first* rejection both Holm and Benjamini–Hochberg use the same threshold α/m that Bonferroni uses everywhere, and the question is whether *any* variable in the set clears the bar. All three therefore give 0 of 56 below 0.25. So does a merely stricter single test at $\alpha = 0.01$ with no multiplicity adjustment at all. And with *no* correction whatsoever, 23 of 56 still fail. The finding depends on there being forty-six tests, not on how one chooses to price them.

The direction of the shortfall is the unfavourable one. The shortest samples belong to the newest variables: the risk-neutral variance measures span under thirty years and face a solo floor of 0.46, nearly twice the effect they would need to establish. A literature that adds variables faster than it adds data moves away from the threshold, not towards it.

6.3 How much of the time was each edge present?

Proposition 4 needs a measured duty cycle to bite. We take it from the certificate's underlying process—not from $\mathcal{E}_t$, which is a running maximum and so rises only when a new high is set—by splitting each evaluation window into ten equal sub-periods and recording the fraction in which log wealth increased. A predictor with no edge scores about one half.

The median across 116 predictors is 0.40, with quartiles 0.30 and 0.50: slightly *below* the one-half baseline a no-edge process produces, which is what a set of variables with no reliable edge looks like. We do not read a duty cycle off the year each wealth path peaks; per Remark 5 that quantity is post-hoc and carries no interval.

Treating 0.40 as a hypothetical ϕ and feeding it through Proposition 4 multiplies the required span by 2.5, equivalently the required ratio by $\sqrt{2.5} = 1.58$. Combined with the corrected floor of 0.39 from Table 7, a variable that was genuinely present only two-fifths of the time would need an on-period annualised information ratio near 0.6—more than twice the threshold for economic significance—for its own sample to have established it. We offer this as an illustration of the direction and magnitude of the intermittency penalty, not as an estimate: the descriptive statistic above is not an estimator of ϕ, and we make no claim that it is.

One degenerate fit, flagged rather than trusted. The variance risk premium at quarterly frequency reports $R^2_{OS} = -98\%$ alongside an e-value of 4.76. This is not a contradiction: its regression forecasts a -78% quarterly equity premium against an outcome standard deviation of 8%, a nine-sigma artifact of a 120-observation fit. Squared-error measures are quadratic and one such point destroys them; the certificate's stake is capped relative to the signal's own scale and is unmoved. We record the largest forecast each fit ever made in units of the outcome's standard deviation and flag anything beyond five. One of 117 is flagged; excluding it changes nothing.

7 Discussion

A null result and a bound are different objects. Welch and Goyal [2] established that these predictors do not beat the historical mean out of sample, and we reproduce it. What that literature has not supplied is the other half: how large an effect the data could have excluded. Without it a negative result is not interpretable, because it is consistent both with there being nothing and with there being something the design could never have seen. Table 1 shows this literature has lived in the second case throughout.

The instrument is not the finding. We are explicit that the certificate is less powerful than Clark–West where Clark–West is well specified [4]. It is used here for three things a t-statistic cannot do: produce a time-uniform ceiling, show where in the sample evidence accumulated, and correct a dependent grid without a bootstrap. Its failure to certify is therefore weak evidence on its own, and the ceilings—which do not depend on rejecting anything—carry the paper.

What would change the answer. Not, within the Gaussian benchmark, a test of the same size and power: Proposition 1 fixes the detectable effect from the span alone. Not finer sampling: Corollary 3. Only a longer span, a larger effect, or a smaller search. Of those, only the last is under a researcher's control, which is an argument for pre-registering a small number of candidates rather than screening many.

8 Limitations

The floor of Proposition 1 assumes a constant effect; a predictor that is strong in some regimes and absent in others has a higher effective floor still, so our numbers are optimistic. Bonferroni is conservative, and a sharper multiplicity correction would lower the corrected column of Table 1, though not to the point of clearing the boundary. The certificate's own conservatism means its ceilings are wider than a fixed-sample interval would be, which cuts against us. And our predictor set is Goyal et al.'s: a set assembled from published papers is selected on past significance, which biases toward finding predictability, not away from it.

9 Conclusion

Evidence about an alpha accumulates as log wealth, so under a Gaussian constant-effect benchmark what a design can detect is fixed in advance by its evaluation span, its size, its target power and the breadth of the search. For the canonical equity-premium data those quantities require 103 years of evaluation for a single prespecified predictor against the effect this literature calls economically significant, and 253 once the search Goyal et al. [1] actually ran is priced. The 1926–2025 data, after a twenty-year burn-in, supply eighty.

Applying an anytime-valid certificate to 117 predictor–frequency pairs certifies nothing, under the primary specification or under any of nine variants. That non-result is the expected one and is not the contribution. The contribution is the interval the certificate supplies without needing to reject: across the 116 unflagged pairs the median marginal 95% upper endpoint on the incremental annualised information ratio is 0.60, and 112 exceed 0.25. A century of the best-studied data in finance leaves most of these predictors compatible both with no predictability at all and with roughly twice as much as would matter. The sequential paths add a description of when each procedure gathered its evidence—never, for the three valuation ratios the literature cites most; late and unevenly for inflation, the Treasury-bill rate and the term spread—which a covariance averaged over the same century cannot show. We report that as heterogeneity in evidence accumulation. Whether it reflects a change in the underlying process is a question this design cannot answer, and we do not claim it does.

A Every predictor

Table 8 reports all 117 predictor–frequency pairs. "Source" records whether the predictor is the real-time diagonal of a Goyal et al. [1] vintage matrix or the workbook column. A dagger marks the single fit flagged as degenerate—a forecast beyond five outcome standard deviations—which is excluded from the summary statistics in the body and changes none of them.

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Table 8: All predictor–frequency pairs, 1926–2025. Rejecting needs an e-value of 20; the largest observed is 7.47. "Ceiling" is the 95% anytime-valid upper bound on the incremental annualised information ratio.

Predictor	Source	n	R^2_{OS} (%)	Clark–West	e-value	Ceiling
Annual						
ogap	workbook	80	3.66	2.54	4.96	1.48
ndrbl	workbook	57	7.30	2.28	3.14	2.17
tail	workbook	80	−0.28	0.29	2.60	0.49
gpce	real time	68	9.53	2.00	2.59	1.47
ltr	workbook	80	−7.67	1.65	2.13	0.53
pce	workbook	62	10.75	3.22	2.13	1.28
lzrt	workbook	80	−2.36	1.46	2.07	0.91
eqis	workbook	80	0.67	1.59	1.93	1.18
d/p	workbook	80	0.22	0.97	1.83	0.76
lty	workbook	80	−6.44	0.94	1.66	0.99
govik	workbook	68	−3.61	1.04	1.64	1.13
i/k	workbook	68	4.89	2.05	1.58	1.25
tbl	workbook	80	−4.59	0.79	1.55	1.09
skvw	workbook	80	−0.12	0.51	1.54	0.79
fbm	workbook	80	−1.26	−1.04	1.46	0.76
d/y	workbook	80	−0.22	1.35	1.37	0.55
d/e	workbook	80	−2.91	−0.69	1.37	0.62
accrul	real time	50	6.68	1.36	1.32	2.12
b/m	workbook	80	−8.12	0.87	1.27	0.80
dfr	workbook	80	−3.95	−0.18	1.26	0.68
e/p	workbook	80	0.95	1.82	1.26	0.59
tchi	workbook	64	−4.83	−0.68	1.25	0.63
svar	workbook	80	−0.18	−0.23	1.16	0.78
sntm	workbook	48	−15.16	0.06	1.15	1.06
wtexas	workbook	80	−4.34	0.81	1.15	1.22
gip	real time	80	−1.86	−1.11	1.13	0.11
disag	workbook	34	−11.30	0.08	1.11	3.67
skew	workbook	59	−2.91	−0.14	1.10	1.13
cfacc	real time	50	7.69	3.34	1.09	2.66
ntis	workbook	80	−6.87	0.27	1.09	1.06
dtoat	workbook	80	−1.33	−0.34	1.05	0.64
cay	workbook	70	−2.99	0.22	1.03	1.34
house	real time	80	−1.04	0.26	1.01	0.38
rdsp	workbook	80	−0.93	0.33	1.01	0.71
dfy	workbook	80	−0.90	−0.76	1.01	0.61
avgcor	workbook	80	−7.10	−0.71	1.01	0.46
dtoy	workbook	80	−6.60	−0.79	1.01	0.43
ygap	workbook	62	−11.46	0.30	1.00	1.07
shtint	workbook	42	−24.60	−1.58	1.00	2.04
csp	workbook	56	−8.65	−0.97	1.00	0.60
infl	workbook	80	−5.93	−1.53	1.00	0.30
tms	workbook	80	−3.15	0.40	1.00	1.07
Monthly						
wtexas	real time	960	0.01	1.85	7.47	0.35
sntm	workbook	641	0.42	2.03	5.92	0.66
ogap	real time	960	0.26	1.73	5.77	0.46
infl	workbook	960	0.11	0.83	5.71	0.51
ndrbl	real time	743	−0.09	0.99	4.57	0.62
csp	workbook	685	−1.05	2.07	4.30	0.46
tms	workbook	960	−0.08	0.78	3.92	0.78
tbl	workbook	960	0.07	1.44	3.48	0.91
ntis	workbook	960	−0.33	0.86	2.84	0.60
disag	real time	468	−0.66	−0.31	2.11	0.99
vrp	real time	360	−6.20	−0.04	2.04	1.31
lzrt	real time	960	−0.17	0.66	1.93	0.40
lty	workbook	960	−0.60	1.38	1.91	0.54
dtoat	real time	960	−0.07	−0.20	1.70	0.62
d/e	workbook	960	−1.52	−0.37	1.47	0.42
d/p	workbook	960	−0.10	1.34	1.45	0.46
dfy	workbook	960	−0.11	−0.75	1.40	0.38
fbm	workbook	960	−1.57	0.17	1.33	0.42
rdsp	real time	960	−0.06	−0.53	1.26	0.44
tchi	real time	779	−0.07	0.51	1.24	0.70
svar	workbook	960	−0.30	−0.51	1.24	0.41
dtoy	real time	960	−0.55	−0.53	1.23	0.43
rsvix	workbook	296	−3.11	−0.81	1.16	0.89
impvar	real time	296	−1.79	0.14	1.15	1.12
d/y	workbook	960	−0.75	1.64	1.14	0.41
tail	real time	960	−0.37	−0.52	1.08	0.36
skvw	real time	960	−0.61	−1.32	1.04	0.18
avgcor	real time	960	−0.17	−0.18	1.02	0.38
vp	real time	324	0.07	0.63	1.02	1.10
ltr	workbook	960	−0.99	−0.56	1.00	0.27
e/p	workbook	960	−0.79	1.55	1.00	0.40
b/m	workbook	960	−1.35	0.59	1.00	0.46
shtint	real time	515	−0.15	0.99	1.00	1.00
ygap	real time	812	−1.02	−0.46	1.00	0.39
dfr	workbook	960	−0.43	−0.23	1.00	0.36
Quarterly						
i/k	workbook	295	2.94	3.19	5.87	0.65
sntm	workbook	213	1.42	2.28	5.33	0.64

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